

SMURF Monitoring of Fish Recruitment at a Shellfish Farm

Protocol ID	underwood_2023_hb221
Sector	Molluscs & echinoderms (blue / green-lipped mussel)
Usability tier	Tier 3 — Partnership (SMURF deployment & retrieval farmer-runnable; ID & analysis partner-supported)
What it monitors	The presence and extent of fish settlement and recruitment at the shellfish farm — the young fish that settle into and recruit to the farm habitat — measured with SMURFs.
Source paper	Underwood, L.H. & Jeffs, A.G. (2023). Settlement and recruitment of fish in mussel farms. <i>Aquaculture Environment Interactions</i> , 15, 85–100.
Associated tools	Part of a bundle: this normalized protocol, a field datasheet, and a stats workbook. Download links are added once the bundle is hosted.

DOI: <https://doi.org/10.3354/aei00454>

Adapted into a general, gear-agnostic method for AquaPIP. Peer-reviewed, open access (CC BY, Aquaculture Environment Interactions).

Introduction

A shellfish farm's suspended structure and biofouling can act as nursery habitat — a place where young fish settle out of the plankton and grow. This protocol measures that directly using SMURFs (standard monitoring units for the recruitment of fish), a simple, standardised collector that samples the settlers and recruits present at the farm.

This paper deployed SMURFs in a New Zealand green-lipped mussel farm and adjacent natural habitats across a settlement season, and found that fish settle and recruit to the farm at rates equivalent to natural reef and that the farm supports growth from settler to juvenile. The authors highlight that SMURFs are a common, reliable, and comparable way to measure recruitment across habitats, and that they are less selective than light traps or diver surveys for cryptic settlers.

Parameters measured

This protocol targets a single indicator — fish settlement and recruitment at the farm. Monitoring linked to it (green) is the SMURF catch; context monitoring (grey) captures the farm setting and biofouling so the recruitment signal can be interpreted and stood behind. Each parameter is listed with its unit and a working definition.

- monitoring linked to a selected indicator
- context monitoring

Fish recruitment — SMURF catch

- **Recruit abundance (count per SMURF)** — the number of fish of each taxon collected from each SMURF.
- **Fish taxon identity** — each fish identified to the finest practical taxonomic level.
- **Size class (total length, mm)** — each fish measured and classed as settler / juvenile / adult (per species size-at-settlement), which distinguishes settlement from later recruitment.
- **Settlement timing** — the month(s) and moon phase at which settlers appear, describing the extent and timing of recruitment at the farm.

Farm setting & biofouling — context monitoring

- **Habitat / depth of SMURF** — where each SMURF sits (e.g. surface vs near-seabed; farm vs a reference habitat) — kept comparable between visits.
- **Biofouling on the farm** — the kelp and other biofouling biomass on the culture lines (from a stripped-section sample), which shapes the habitat structure.

- **Water conditions** — tidal range, currents, and season at the site.

Requirements

- **Personnel [estimate]:** a boat crew with a crane/winch to deploy and lift the SMURFs (farmer-runnable); and someone (or a partner) able to identify and size-class the fish and run the analysis.
- **Time per visit [estimate]:** a sampling session each month, timed to the new moon — lift each SMURF, remove and process the fish, redeploy; plus periodic partner analysis.

Sampling frequency & timing

- **Framework guidance.** The MEL framework sets the mollusc spawning/recruitment indicator (2.2.1) at seasonal, timed to spawning and recruitment, sampling within the farm.
- **What the paper reports.** This paper samples monthly across a ~3-month settlement season (summer), with fish removed from each SMURF around the new moon.
- **Recommended to detect an effect.** Sample monthly through the local settlement season (timed to the new moon, when recruitment peaks), and repeat across [2–3+] seasons so year-to-year variability in larval supply can be separated from a farm signal.

► A reference habitat helps interpretation

Recruitment is measured within the farm (the framework samples this indicator “within the farm”, so no comparison is strictly required). But deploying matched SMURFs in a nearby natural habitat — as the source did — shows whether the farm recruits fish like natural habitat does. Reference-habitat choice is [bracketed].

Materials

Materials are roles to fill, not products to buy. SMURFs are inexpensive and standardised; the identification is the specialist part.

► Gear varies site to site — use your best judgement

The source built SMURFs to a published specification and used specific software. Substitute equivalent recruitment collectors, a boat/crane, and any stats package you have. The specialist need is fish identification and size-classing. Nothing below is mandatory as a specific product.

- **For the survey:** SMURFs (standardised plastic-mesh recruitment collectors, per a published spec); mooring lines with anchors/buoys, or ties to the farm backbone; a boat with a crane/winch; a large scoop net [e.g. 5 mm mesh]; a fish bin; a measuring board (total length) and a reference scale for photos; a balance; fish ID references; and a stats package.

Methods

Single-method protocol — deploy SMURFs at the farm, sample them monthly through the settlement season, and process the fish. Deployment and retrieval are farmer-runnable; identification, size-classing, and analysis are partner-supported (see tier).

Deploy and sample SMURFs

↳ *Linked indicator: supports spawning/recruitment — fish settling and recruiting at the farm — match: **Exact**. Full mapping in Appendix A.*

Materials. SMURFs; mooring lines / backbone ties; a boat with a crane/winch; a large scoop net; a fish bin; a measuring board + reference scale; a balance; fish ID references; a stats package.

- **Preparation.** Build or obtain SMURFs to a standard specification. Fix the number of SMURFs [e.g. 10] and their positions — spaced apart [e.g. ≥ 30 m], at set depths [e.g. a surface and a near-seabed set]. Optionally place matched SMURFs in a nearby natural reference habitat.
- **Deployment.** Moor the SMURFs at the farm (tied to the backbone) and at any reference habitat. Log each SMURF's position and depth.
- **Sampling.** Around the new moon each month through the settlement season, lift each SMURF gently to the surface, enclose it in a scoop net to retain fish, and flush the fish into a bin. Redeploy the SMURF to the same position.
- **Records & analysis.** Identify, count, measure (total length), and size-class each fish (settler / juvenile / adult per species). Summarise recruit abundance, taxa richness, a diversity index, and the timing of settlement per SMURF and month. A simple summary answers the indicator; the source's linear mixed models and NMDS comparison across habitats are the partner-supported statistical steps.

Expect these failure modes and constraints (and manage them)

Each constraint is grouped under the method or stage it affects, with how to manage it.

SMURF sampling

- **Species-specific retention** — some species associate with SMURFs only briefly, or move in and out during the post-settlement period; treat SMURF counts as a comparable relative index of recruitment, not an absolute census.
- **Retrieval loss** — fish can escape as a SMURF is lifted; enclose it in the scoop net as it nears the surface (fish retreat into the collector, so loss is usually small) and keep the method constant.
- **Settler-vs-recruit identification** — distinguishing settlers from older recruits needs size-at-settlement for each species; fix the size classes up front and record the rule.

Across design & interpretation

- **Larval-supply variability** — recruitment depends on larval supply, which varies by year and location; a matched reference habitat and repetition across seasons separate a farm effect from background variation.
- **Single site / short season** — one farm over one season captures a snapshot; repeat across seasons before drawing firm conclusions.

Data handling

Use a consistent labelling convention: site (farm / reference) — SMURF — depth — date (moon phase). Enter counts, lengths, and size classes into the companion workbook; its data-entry columns mirror the datasheet exactly. Keep the farm and reference records clearly separated. [Insert your own site photo] where useful.

Methods referenced from other sources

The core collector follows an established design. Resolve and verify a stable link before relying on it.

- **SMURF design & recruitment sampling** — the construction and use of SMURFs in the Preparation and Sampling steps. Ammann, A.J. (2004). SMURFs: standard monitoring units for the recruitment of temperate reef fishes. *Journal of Experimental Marine Biology and Ecology*, 299, 135–154. [[resolve stable link](#) — [verify](#)]

Appendix A — MEL objective table

Reference: this protocol targets one indicator; the row below is the full form of the per-method “Linked indicator” line above. Framework codes appear only here.

Goal	Monitoring method	MEL Objective	Metric(s)	Match
GOAL 2 Molluscs and echinoderms are farmed in a way that enhances the abundance of associated native fauna by supporting spawning or recruitment.	SMURFs (standard monitoring units for the recruitment of fish)	Objective 2.2 (Ind. 2.2.1) — Farming enhances abundance by supporting spawning or recruitment; species using and extent of recruitment at the farm site	Presence and extent of recruitment — settlers and recruits per SMURF, by taxon and size class, over the settlement season	 Exact

Match scale. How closely the method's metric matches the objective's.

-  **Exact** — measures the objective's metric as specified.
 **Partial** — measures a proxy for, or subset of, the metric, or needs an added condition such as a control site.
 **Indirect** — gives relevant evidence toward the objective (a native species present/active) but not the specified metric.

Not mapped (context). The SMURF habitat/depth, on-farm biofouling, and water conditions are recorded to interpret the recruitment signal but are not themselves mapped. SMURFs measure recruits specifically — the general mobile-fauna assemblage and a broader community comparison are not covered here (mobile fauna is covered by ferriss_2021_hb121).

Appendix B — Provenance & practitioner decisions

- **Adapted from:** Underwood, L.H. & Jeffs, A.G. (2023). Settlement and recruitment of fish in mussel farms. *Aquaculture Environment Interactions*, 15, 85–100. <https://doi.org/10.3354/aei00454> (peer-reviewed; open access, CC BY).
- **Indicator set:** This protocol targets a single indicator — spawning/recruitment (match Exact; SMURFs are the framework's suggested method for 2.2.1, measuring the extent of recruitment within the farm). Tier T3: SMURF deployment and retrieval are farmer-runnable, but fish identification, size-classing, and analysis need partner support. Fills the mollusc recruitment gap left by removing the non-peer-reviewed hossain_2013. The source also reports fish abundance and a community comparison, but those are recruit-specific and do not robustly map to the mobile-fauna or community-composition indicators (out of scope here).
- **Decisions left to the practitioner [bracketed]:** these are choices this protocol leaves open for the site — not gaps the authors reported. Set them locally: number of SMURFs, their depths, and spacing; whether matched reference-habitat SMURFs are deployed; the local settlement season and sampling months; the size-at-settlement thresholds for each species; and whether the optional statistical comparison is run.